Function

36. Let A be the set of all function $f : Z \to Z$ and R be a ralation on A such that

$R = \{(f, g) : f(0) = g(1) \text{ and } f(1) = g(0)\}$. Then R is :

(A) Reflexive but neither symmetric nor transitive

(B) Symmetric but neither reflecxive nor transitive

(C) Transitive but neither reflexive nor symmetric

(D) Symmetric and transitive but not reflective

माना सभी फलनों $f : Z \to Z$ का समुच्चय A है और A पर एक संबंध $R = \{(f, g) : f(0) = g(1)$ और $f(1) = g(0)$ हैं$\}$ है। तो R होगा :

(A) स्वतुल्य है किन्तु न तो सममित है न ही संक्रामक है।

(B) सममित है किन्तु न तो स्वतुल्य है न ही संक्रामक है।

(C) संक्रामक है किन्तु न तो स्वतुल्य है न ही सममित है।

(D) सममित और संक्रामक है किन्तु स्वतुल्य नहीं है।

Ans. B

Sol. $R = \{(f, g) : f(0) = g(1) \text{ and } f(1) = g(0)\}$

Reflexive: $(f, f) \in R$

$= f(0) = f(1)$ and $f(1) = f(0) \to$ must hold

$\Rightarrow$ but this is not true for all function so not reflexive

Symmetric: If $(f, g) \in R \Rightarrow (g, f) \in R$

Now, $g(0) = f(1)$ and $g(1) = f(0) \to$ true

$\therefore$ symmetric

Transitive : If$(f, g) \in R$ and $(g, h) \in R$

$\Rightarrow (f, h) \in R$

Now $(f, g) \in R \Rightarrow f(0) = g(1)$ and $f(1) = g(0)$

$(g, h) \in R \Rightarrow g(0) = h(1)$ and $g(1) = h(0)$

For $(f, h) \in R$ we need $f(0) = h(1)$ and $f(1) = h(0)$

Now $f(0) = g(1) = h(0)$ and $f(1) = g(0) = h(1)$

Hence not transitive

37. If the domain of the function $f(x)=\frac{1}{\sqrt{10+3x-x^2}}+\frac{1}{\sqrt{x+|x|}}$ is (a, b), then $(1+a)^2+b^2$ is equal to :

यदि फलन $f(x)=\frac{1}{\sqrt{10+3x-x^2}}+\frac{1}{\sqrt{x+|x|}}$ का प्रांत (a, b) है, तो $(1 + a)^2 + b^2$ बराबर है :

(A) 26 (B) 29 (C) 30 (D) 25

Ans. A

Sol. $x+|x|=\begin{cases}2x, & x\geq 0\\ 0, & x<0\end{cases}$

$\Rightarrow \frac{1}{\sqrt{x+|x|}}$, domain is $x > 0$, as $2x \neq 0$

Similarly,

$\frac{1}{\sqrt{3x+10-x^2}}$ is defined when $3x + 10 - x^2 > 0$

$\Rightarrow x^2 - 3x - 10 < 0$

$(x - 5)(x + 2) < 0$

$\Rightarrow x \in (-2, 5)$

$\Rightarrow$ Domain will be $(0, \infty) \cap (-2, 5) = (0, 5)$

$\Rightarrow (1 + a)^2 + b^2 = 1 + 25 = 26$

38. Let f be a function such that $f(x)+3f\left(\frac{24}{x}\right)=4x$, $x \neq 0$. Then f(3) + f(8) is equal to :

माना एक फलन f के लिए $f(x)+3f\left(\frac{24}{x}\right)=4x$, $x \neq 0$ है। तो f(3) + f(8) बराबर है :

(A) 10 (B) 12 (C) 11 (D) 13

Ans. C

Sol. $f(x)+3f\left(\frac{24}{x}\right)=4x$

Put x = 3	$f(3) + 3f(8) = 12$
Put x = 8	$f(8) + 3f(3) = 32$
Add both	$4(f(3) + f(8)) = 44$
	$f(3) + f(8) = 11$

39. If the domain of the function $f(x) = \log_7 (1 - \log_4(x^2 - 9x + 18))$ is $(\alpha, \beta) \cup (\gamma, \delta)$, then $\alpha + \beta + \gamma + \delta$ is equal to :

यदि फलन $f(x) = \log_7 (1 - \log_4(x^2 - 9x + 18))$ का प्रांत $(\alpha, \beta) \cup (\gamma, \delta)$ है, तो $\alpha + \beta + \gamma + \delta$ बराबर है :

(A) 16 (B) 18 (C) 15 (D) 17

Ans. B

Sol. $1 - \log_4 (x^2 - 9x + 18) > 0$

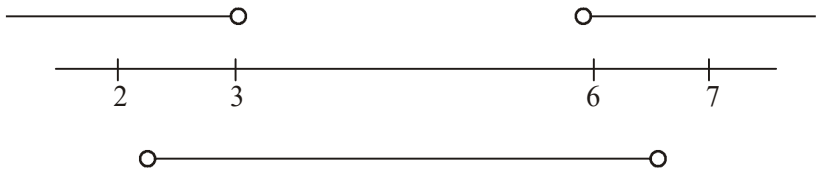
$\Rightarrow \log_4 (x^2 - 9x + 18) < 1$

$\Rightarrow 0 < x^2 - 9x + 18 < 4$

$x^2 - 9x + 18 > 0$ & $x^2 - 9x + 18 < 4$

$(x - 3)(x - 6) > 0$ & $x^2 - 9x + 14 < 0$

$(x - 2)(x - 7) < 0$



$(\alpha, \beta) \cup (\gamma, \delta) = (2, 3) \cup (6, 7)$

So, $\alpha + \beta + \gamma + \delta = 5 + 13 = 18$

40. Let f, g : $(1, \infty) \to R$ be defined as $f(x) = \frac{2x+3}{5x+2}$ and $g(x) = \frac{2-3x}{1-x}$. If the range of the function $fog : [2, 4] \to R$ is $[\alpha, \beta]$, then $\frac{1}{\beta - \alpha}$ is equal to :

माना f, g : $(1, \infty) \to R$, $f(x) = \frac{2x+3}{5x+2}$ और $g(x) = \frac{2-3x}{1-x}$ द्वारा परिभाषित हैं। यदि फलन $fog : [2, 4] \to R$ का परिसर $[\alpha, \beta]$ है, तो $\frac{1}{\beta - \alpha}$ बराबर है :

(A) 29 (B) 2 (C) 56 (D) 68

Ans. C

Sol. $f, g : (1, \infty) \to R$

$f(x) = \frac{2x+3}{5x+2}$; $g(x) = \frac{2-3x}{1-x}$

$f(g(x)) = \frac{4 - 6x + 3 - 3x}{10 - 15x + 2 - 2x}$

$= \frac{7 - 9x}{12 - 17x}$

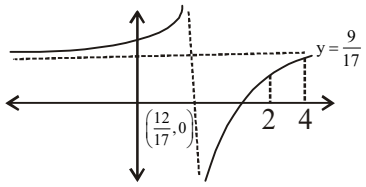
$f(g(x)) : [2, 4] \to R$

$f(g(x)) = \frac{2(g(x)) + 3}{5(g(x)) + 2}$

$$= \frac{2\left\{\frac{2-3x}{1-x}\right\}+3}{5\left\{\frac{2-3x}{1-x}\right\}+2}$$

$f(g(x)) = \frac{9x-7}{17x-12} \rightarrow$ Range : $R - \left\{\frac{9}{17}\right\}$

But its domain is [2, 4]



$f(g(2)) = \frac{9.2-7}{17.2-12} = \frac{1}{2}$

$f(g(4)) = \frac{9.4-7}{17.4-12} = \frac{29}{56}$

$[\alpha, \beta] = \left[\frac{1}{2}, \frac{29}{56}\right]$ then

$\beta - \alpha = \frac{29}{56} - \frac{1}{2}$ $\qquad \frac{1}{\beta-\alpha} = 56$

$= \frac{29-28}{56}$

$= \frac{1}{56}$

41. Consider the sets A = {(x, y) ∈R × R : $x^2 + y^2 = 25$}, B = {(x, y) ∈ R × R : $x^2 + 9y^2 = 144$}, C = {(x, y) ∈ Z × Z : $x^2 + y^2 \le 4$} and D = A ∩ B. The total number of one-one functions from the set D to the set C is :

समुच्चयों A = {(x, y) ∈R × R : $x^2 + y^2 = 25$}, B = {(x, y) ∈ R × R : $x^2 + 9y^2 = 144$}, C = {(x, y) ∈ Z × Z : $x^2 + y^2 \le 4$} और D = A ∩ B का विचार कीजिए। समुच्चय D से समुच्चय C के एकैकी फलनों की कुल संख्या है :

(A) 15120 (B) 19320 (C) 18290 (D) 17160

Ans. D

Sol. $A \equiv x^2 + y^2 = 25$

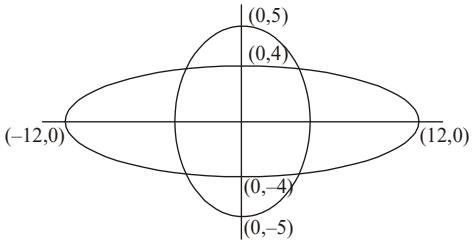
$B \equiv x^2 + 9y^2 = 144$

$C \equiv x^2 + y^2 \le 4$: $x, y \in z \times z$

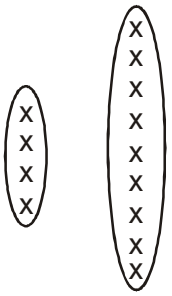
$\therefore$ {(2,0), (0,2), (–2,0), (0,–2)

(1, 0) (0, 1) (–1,0) (0,–1)

(1,1) (1,–1) (–1,1) (–1,–1), (0,0)}



$\therefore$ number of elements in $n(A \cap B) = 4$ & $n(C) = 13$



Number of one-one function

$= {}^{13}C_4 \times 4!$

$= \frac{13!}{9!4!} \times 4!$

$= 13 \times 12 \times 11 \times 10$

$\Rightarrow 17160$

42. Let A={1, 2, 3, 4} and B = {1, 4, 9, 16}. Then the number of many-one functions f: A $\to$ B such that $1 \in f(A)$ is equal to :

माना कि A={1, 2, 3, 4} और B = {1, 4, 9, 16} है। तब बहुएकैकी फलनों f: A $\to$ B की संख्या, इस प्रकार है कि $1 \in f(A)$ है, बराबर है :

(A) 139 (B) 151 (C) 163 (D) 127

Ans. B

Ans. Total function – (one – one) – (B – {1})

$4^4 - 4! - 3^4$

$= 151$

43. Let the range of the function $f(x) = 6 + 16\cos x \cdot \cos\left(\frac{\pi}{3} - x\right) \cdot \cos\left(\frac{\pi}{3} + x\right) \cdot \sin 3x \cdot \cos 6x$, $x \in R$ be $[\alpha, \beta]$. Then the distance of the point (α, β) from the line $3x + 4y + 12 = 0$ is :

माना फलन $f(x) = 6 + 16\cos x \cdot \cos\left(\frac{\pi}{3} - x\right) \cdot \cos\left(\frac{\pi}{3} + x\right) \cdot \sin 3x \cdot \cos 6x$, $x \in R$ का परिसर $[\alpha, \beta]$ है। तब बिन्दु (α, β) की रेखा $3x + 4y + 12 = 0$ से दूरी है :

(A) 8 (B) 11 (C) 9 (D) 10

Ans. B

Sol. $f(x)=6+16\cos x\cdot\cos\left(\frac{\pi}{3}-x\right)\cdot\cos\left(\frac{\pi}{3}+x\right)\cdot\sin 3x\cdot\cos 6x$

$=6+16\cos x\left[\cos^2\frac{\pi}{3}-\sin^2 x\right]\sin 3x\cos 6x$

$=6+16\cos x\left[\frac{1}{4}-\sin^2 x\right]\sin 3x\cos 6x$

$=6+4\cos x\,(1-4\sin^2 x)\sin 3x\cos 6x$

$\downarrow$

$(1-\cos^2 x)$

$=6+4\cos x\,(4\cos^2 x-3)\sin 3x\cos 6x$

$=6+4[4\cos^3 x-3\cos x]\sin 3x\cos 6x$

$=6+4\cos 3x\sin 3x\cos 6x$

$=6+\sin 12x$

$-1\le\sin 12x\le 1$

$5\le 6+\sin 12x\le 7$

$\therefore(\alpha,\beta)\equiv[5,7]$

$d=\frac{|3.5+4.7+12|}{5}$

$d=\frac{|15+28+12|}{5}$

$=11$

44. Let $f(x)=\frac{2^{x+2}+16}{2^{2x+1}+2^{x+4}+32}$. Then the value of $8\left(f\left(\frac{1}{15}\right)+f\left(\frac{2}{15}\right)+\ldots+f\left(\frac{59}{15}\right)\right)$ is equal to :

माना कि $f(x)=\frac{2^{x+2}+16}{2^{2x+1}+2^{x+4}+32}$ है। तब $8\left(f\left(\frac{1}{15}\right)+f\left(\frac{2}{15}\right)+\ldots+f\left(\frac{59}{15}\right)\right)$ का मान बराबर है :

(A) 108 (B) 102 (C) 118 (D) 92

Ans. C

Sol. $f(x)=\frac{42^x+16}{2.2^{2x}+16.2^x+32}$

$f(x)=\frac{2\left(2^x+4\right)}{2^{2x}+8.2^x+16}$

$f(x)=\frac{2}{2^x+4}$

$$f(4-x)=\frac{2^x}{2(2^x+4)}$$

$$f(x)+f(4-x)=\frac{1}{2}$$

So, $f\left(\frac{1}{15}\right)+f\left(\frac{59}{15}\right)=\frac{1}{2}$

Similarly $=f\left(\frac{29}{15}\right)+f\left(\frac{31}{15}\right)=\frac{1}{2}$

$$f\left(\frac{30}{15}\right)=f(2)=\frac{2}{2^2+4}=\frac{2}{8}=\frac{1}{4}$$

$$\Rightarrow 8\left(29\times\frac{1}{2}+\frac{1}{4}\right)$$

Ans. 118

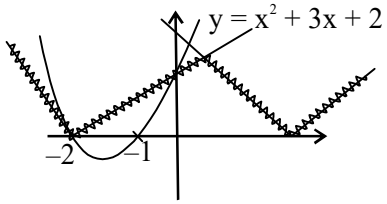
45. The number of real solution(s) of the equation $x^2+3x+2=\min\{|x-3|,|x+2|\}$ is :

समीकरण $x^2+3x+2=\min\{|x-3|,|x+2|\}$ के वास्तविक हल/हलों की संख्या है :

(A) 2 (B) 0 (C) 3 (D) 1

Ans. A

Sol.



2 Solutions

46. The function $f:(-\infty,\infty)\to(-\infty,1)$, defined by $f(x)=\frac{2^x-2^{-x}}{2^x+2^{-x}}$ is :

(A) Both one-one and onto (B) Neither one-one nor onto

(C) One-one but not onto (D) Onto but not one-one

$f(x)=\frac{2^x-2^{-x}}{2^x+2^{-x}}$ द्वारा परिभाषित फलन $f:(-\infty,\infty)\to(-\infty,1)$:

(A) एकैकी और आच्छादक दोनों है (B) ना तो एकैकी है और ना ही आच्छादक है

(C) एकैकी है परन्तु आच्छादक नहीं (D) आच्छादक है परंतु एकैकी नहीं है

Ans. C

Sol. $f(x)=\frac{2^{2x}-1}{2^{2x}+1}=1-\frac{2}{2^{2x}+1}\in(-1,1)$

$f'(x)=\frac{2}{\left(2^{2x}+1\right)^2}\cdot 2^{2x}\cdot 2\cdot \ell n2>0$

f(x) is one-one but not onto

47. If $f(x)=\frac{2^x}{2^x+\sqrt{2}}, x\in\mathbb{R}$, then $\sum_{k=1}^{81} f\left(\frac{k}{82}\right)$ is equal to :

यदि $f(x)=\frac{2^x}{2^x+\sqrt{2}}, x\in\mathbb{R}$ है, तो $\sum_{k=1}^{81} f\left(\frac{k}{82}\right)$ बराबर है :

(A) 82 (B) $\frac{81}{2}$ (C) $81\sqrt{2}$ (D) 41

Ans. B

Sol. $f(x)=\frac{2^x}{2^x+\sqrt{2}}$

$f(x)+f(1-x)=\frac{2^x}{2^x+\sqrt{2}}+\frac{2^{1-x}}{2^{1-x}+\sqrt{2}}$

$=\frac{2^x}{2^x+\sqrt{2}}+\frac{2}{2+\sqrt{2}2^x}=\frac{2^x+\sqrt{2}}{2^x+\sqrt{2}}=1$

Now, $\sum_{k=1}^{81} f\left(\frac{k}{82}\right)=f\left(\frac{1}{82}\right)+f\left(\frac{2}{82}\right)+\ldots\ldots+f\left(\frac{81}{82}\right)$

$=f\left(\frac{1}{82}\right)+f\left(\frac{1}{82}\right)+\ldots\ldots..f\left(1-\frac{2}{82}\right)+f\left(1-\frac{1}{82}\right)$

$\left[f\left(\frac{1}{82}\right)+f\left(1-\frac{1}{82}\right)\right]+\left[f\left(\frac{2}{82}\right)+f\left(1-\frac{2}{82}\right)\right]+\ldots.40 \text{ cases } +f\left(\frac{41}{82}\right)$

$(1+1+\ldots..+1)40 \text{ times } +\frac{2^{\frac{1}{2}}}{2^{\frac{1}{2}}+2^{\frac{1}{2}}}$

$40+\frac{1}{2}=\frac{81}{2}$

48. Let $f: R-\{0\}\to(-\infty, 1)$ be a polynomial of degree 2, satisfying $f(x)f\left(\frac{1}{x}\right)=f(x)+f\left(\frac{1}{x}\right)$. If f(K) = –2 K,

then the sum of squares of all possible values of K is :

माना $f : R - \{0\} \to (-\infty, 1)$ घात 2 का एक बहुपद है, जो कि $f(x)f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$ को संतुष्ट करता है। यदि $f(K) = -2K$, तो K के सभी संभव मानों के वर्गों का योग है :

(A) 9 (B) 7 (C) 1 (D) 6

Ans. D

Sol. $f(x) = (1 - x^2)$

$f(k) = (1 - k^2) = -2k$

$k^2 - 2k - 1 = 0 \begin{cases} \alpha \\ \beta \end{cases}$

$(\alpha^2 + \beta^2) = (\alpha + \beta)^2 - 2\alpha\beta$

$\Rightarrow (2)^2 - 2(-1)$

$\Rightarrow 4 + 2 = 6$

49. Let $f : [0,3] \to A$ be defined by $f(x) = 2x^3 - 15x^2 + 36x + 7$ and $g : [0, \infty) \to B$ be defined by $g(x) = \frac{x^{2025}}{x^{2025} + 1}$. If both the functions are onto and $S = \{x \in Z : x \in A \text{ or } x \in B\}$, then n(S) is equal to :

माना कि $f : [0,3] \to A$, $f(x) = 2x^3 - 15x^2 + 36x + 7$ द्वारा परिभाषित होता है और $g : [0, \infty) \to B$, $g(x) = \frac{x^{2025}}{x^{2025} + 1}$ द्वारा परिभाषित होता है। यदि दोनों फलन आच्छादक हैं और $S = \{x \in Z : x \in A$ या $x \in B\}$ है, तो n(S) बराबर है :

(A) 30 (B) 31 (C) 36 (D) 29

Ans. A

Sol. $S = \{x \in Z \text{ (Integers)} : x \in A \text{ or } x \in B\}$

$\Rightarrow$ All Integer values in set A & set B.

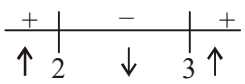
(i) $f : [0, 3] \to A$

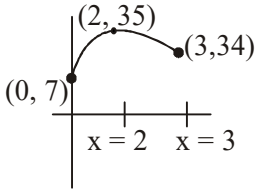
$f(x) = 2x^3 - 15x^2 + 36x + 7$

$f'(x) = 6x^2 - 30x + 36$

$\Rightarrow 6(x^2 - 5x + 6)$

$\Rightarrow 6(x - 3)(x - 2)$





$f(2) = 2(8) - 15(4) + 36(2) + 7$

$\Rightarrow 16 - 60 + 72 + 7$

$\Rightarrow (28 + 7) = 35$

$f(3) = 2(27) - 15(9) + 36(3) + 7$

$\Rightarrow 54 - 135 + 108 + 7$

$\Rightarrow 34$

$A \Rightarrow [7, 35] \Rightarrow$ Total integers $\rightarrow 29$

(ii) $g ; [0, \infty) \rightarrow B$

$$g(x) = \left(\frac{x^{2025}}{x^{2025}+1}\right)$$

$$g(x) = 1 - \frac{1}{\left(x^{2025}+1\right)}$$

$B \Rightarrow [0, 1) \Rightarrow$ Total integer = 1
$n(S) = A + B \Rightarrow (29 + 1) = 30$ Integers

50. Let $A = \{1, 3, 7, 9, 11\}$ and $B = \{2, 4, 5, 7, 8, 10, 12\}$. Then the total number of one-one maps $f: A \rightarrow B$, such that $f(1) + f(3) = 14$, is :

माना $A = \{1, 3, 7, 9, 11\}$ तथा $B = \{2, 4, 5, 7, 8, 10, 12\}$ हैं। तो एकैकी प्रतिचित्रों (maps) $f: A \rightarrow B$, जिनके लिए $f(1) + f(3) = 14$ है, की कुल संख्या है :

(A) 180 (B) 120 (C) 480 (D) 240

Ans. D

Sol. $A = \{1, 3, 7, 9, 11\}$

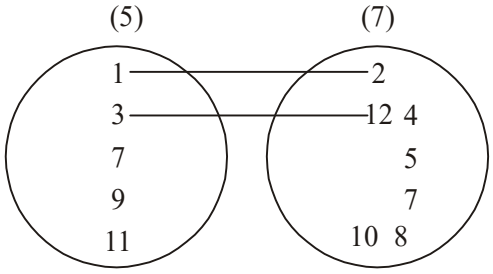
$B = \{2, 4, 5, 7, 8, 10, 12\}$

$f(1) + f(3) = 14$

(i) 2 + 12

(ii) 4 + 10

$2 \times (2 \times 5 \times 4 \times 3) = 240$



51. Let $f, g : R \rightarrow R$ be defined as :

$$f(x) = |x - 1| \text{ and } g(x) = \begin{cases} e^x, & x \geq 0 \\ x + 1, & x \leq 0 \end{cases}$$

Then the function $f(g(x))$ is :

(A) Neither one-one nor onto (B) Onto but not one-one

(C) One-one but not onto (D) Both one-one and onto

माना $f, g : R \to R$,

$f(x) = |x-1|$ तथा $g(x) = \begin{cases} e^x, & x \geq 0 \\ x+1, & x \leq 0 \end{cases}$

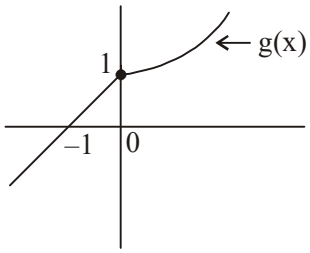
द्वारा परिभाषित हैं। तो फलन $f(g(x))$:

(A) न तो एकैकी न ही आच्छादक है (B) आच्छादक है परन्तु एकैकी नहीं है

(C) एकैकी है परन्तु आच्छादक नहीं है (D) एकैकी तथा आच्छादक दोनों है

Ans. A

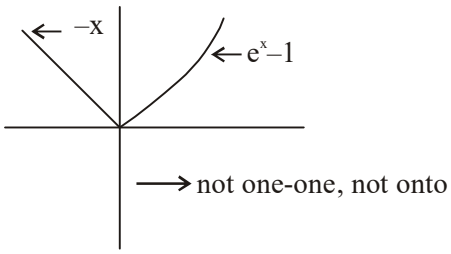
Sol. $f(x) = \begin{cases} x-1 & x \geq 1 \\ 1-x & x < 1 \end{cases}$



$$g(x) = \begin{cases} e^x & ; \quad x \geq 0 \\ x+1 & ; \quad x \leq 0 \end{cases}$$

$$f(g(x)) = \begin{cases} g(x)-1, & g(x) \geq 1 \\ 1-g(x) & g(x) < 1 \end{cases}$$

$$= \begin{cases} e^x - 1 & x \geq 0 \\ -x & x < 0 \end{cases}$$



52. The function $f : N - \{1\} \to N$; is defined by f(n) = the highest prime factor of n, is :

(A) neither one-one nor onto (B) onto only

(C) one-one only (D) both one-one and onto

फलन $f : N - \{1\} \to N$; f(n) = पूर्णांक n का सबसे बड़ा अभाज्य गुणनखंड, द्वारा परिभाषित है। तो f :

(A) न तो एकैकी है न ही आच्छादक है (B) केवल आच्छादक है

(C) केवल एकैकी है (D) एकैकी और आच्छादक दोनों है

Ans. A

Sol. Largest prime factors of two different natural numbers can be same so it is many one.

For example $f(6) = 3, f(12) = 3$

$f(10) = 5, f(15) = 5$

As range of f(x) is set of prime number so range of f(x) is proper subset of co-domain so f is into.

53. If $f(x) = \begin{cases} 2+2x, & -1 \le x < 0 \\ 1-\frac{x}{3}, & 0 \le x \le 3 \end{cases}$; $g(x) = \begin{cases} -x, & -3 \le x \le 0 \\ x, & 0 < x \le 1 \end{cases}$, then range of (fog) (x) is :

यदि $f(x) = \begin{cases} 2+2x, & -1 \le x < 0 \\ 1-\frac{x}{3}, & 0 \le x \le 3 \end{cases}$; $g(x) = \begin{cases} -x, & -3 \le x \le 0 \\ x, & 0 < x \le 1 \end{cases}$, तो (fog) (x) का परिसर है :

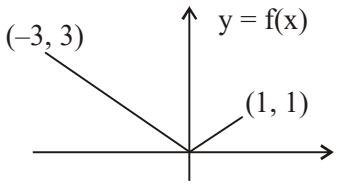
(A) [0, 1) (B) [0, 3) (C) (0, 1] (D) [0, 1]

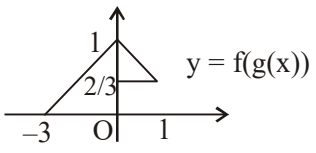
Ans. D

Sol. $f(g(x)) = \begin{cases} 2+2g(x) & , \quad -3 \le g(x) < 0 \quad ____(1) \\ 1-\frac{g(x)}{3} & , \quad 0 \le g(x) \le 3 \quad ____(2) \end{cases}$

By (1) $x \in \phi$

And by (2) $x \in [-3, 0]$ and $x \in [0, 1]$





Range of f(g(x)) is [0, 1]

54. If $f(x)=\frac{4x+3}{6x-4}$, $x\neq\frac{2}{3}$ and (fof) (x) = g(x), where $g: R-\left\{\frac{2}{3}\right\}\to R-\left\{\frac{2}{3}\right\}$, then (gogog) (4) is equal to :

यदि $f(x)=\frac{4x+3}{6x-4}$, $x\neq\frac{2}{3}$ तथा (fof) (x) = g(x) है, जहाँ $g: R-\left\{\frac{2}{3}\right\}\to R-\left\{\frac{2}{3}\right\}$ है, तो (gogog) (4) बराबर है:

(A) –4 (B) $-\frac{19}{20}$ (C) $\frac{19}{20}$ (D) 4

Ans. D

Sol. $g(x)=f\left(\frac{4x+3}{6x-4}\right)=\frac{4\left(\frac{4x+3}{6x-4}\right)+3}{6\left(\frac{4x+3}{6x-4}\right)-4}=\frac{16x+12+18x-12}{24x+18-24x+16}=\frac{34x}{34}=x$

gogog(4) = g(g(4)) = g(4) = 4

55. Let $f: R\to R$ and $g: R\to R$ be defined as $f(x)=\begin{cases}\log_e x & , \quad x>0\\ e^{-x} & , \quad x\le 0\end{cases}$ and $g(x)=\begin{cases}x & , \quad x\ge 0\\ e^{x} & , \quad x<0\end{cases}$. Then

$gof: R\to R$ is :

(A) both one-one and onto (B) onto but not one-one

(C) one-one but not onto (D) neither one-one nor onto

माना $f: R\to R$ तथा $g: R\to R$, $f(x)=\begin{cases}\log_e x & , \quad x>0\\ e^{-x} & , \quad x\le 0\end{cases}$ तथा $g(x)=\begin{cases}x & , \quad x\ge 0\\ e^{x} & , \quad x<0\end{cases}$ द्वारा परिभाषित है। तो

$gof: R\to R$:

(A) एकैकी एवं आच्छादक दोनों है (B) आच्छादक है परन्तु एकैकी नहीं है

(C) एकैकी है परन्तु आच्छादक नहीं है (D) न तो एकैकी है, न ही आच्छादक है

Ans. D

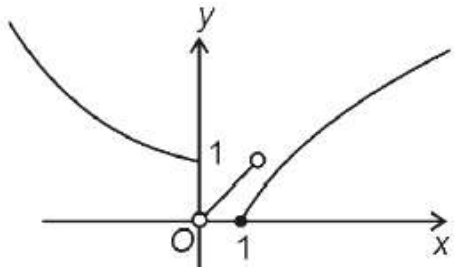
Sol. Given $f(x)=\begin{cases}e^{-x} & ; \quad x\le 0\\ \ell nx & ; \quad x>0\end{cases}$

and $g(x)=\begin{cases}x & ; \quad x\ge 0\\ e^{x} & ; \quad x<0\end{cases}$

then gof(x)
= g(f(x))

$$= \begin{cases} f(x) & ; \quad f(x) \geq 0 \\ e^{f(x)} & ; \quad f(x) < 0 \end{cases}$$

$$= \begin{cases} e^{-x} & ; \quad x \leq 0 \\ x & ; \quad 0 < x < 1 \\ \ell nx & ; \quad x \geq 1 \end{cases}$$



$gof(x)$

$\therefore$ f(x) is neither one-one nor onto

56. Let $A = \{x \in \mathbb{R} : [x+3]+[x+4] \leq 3\}$, $B = \left\{x \in \mathbb{R} : 3^x \left(\sum_{r=1}^{\infty} \frac{3}{10^r}\right)^{x-3} < 3^{-3x}\right\}$, where [t] denotes greatest integer function. Then :

माना $A = \{x \in \mathbb{R} : [x+3]+[x+4] \leq 3\}$, $B = \left\{x \in \mathbb{R} : 3^x \left(\sum_{r=1}^{\infty} \frac{3}{10^r}\right)^{x-3} < 3^{-3x}\right\}$ है, जहाँ [t] महत्तम पूर्णांक फलन है। तब :

(A) $A = B$ (B) $B \subset C, A \neq B$ (C) $A \cap B = \phi$ (D) $A \subset B, \quad A \neq B$

Ans. A

Sol. $[x]+3+[x]+4 \leq 3$

$2[x] \leq -4$

$[x] \leq -2 \Rightarrow x \in (-\infty, -1)$. _____A

$$3^x \left(\frac{3 \cdot \frac{1}{10}}{1-\frac{1}{10}}\right)^{x-3} < 3^{-3x}$$

$27 < 3^{-3x}$

$-3x > +3$

$x < -1$ _____B

$A = B$

57. If $f(x)=\frac{\left(\tan 1^{\circ}\right)x+\log_e(123)}{x\log_e(1234)-\left(\tan 1^{\circ}\right)}, x>0$, then the least value of $f(f(x))+f\left(f\left(\frac{4}{x}\right)\right)$ is :

यदि $f(x)=\frac{\left(\tan 1^{\circ}\right)x+\log_e(123)}{x\log_e(1234)-\left(\tan 1^{\circ}\right)}, x>0$ है, तो $f(f(x))+f\left(f\left(\frac{4}{x}\right)\right)$ का निम्नतम मान है :

(A) 2 (B) 8 (C) 4 (D) 0

Ans. C

Sol. $f(x)=\frac{(\tan 1)x+\log 123}{x\log 1234-\tan 1}$

Let $A=\tan 1, B=\log 123, C=\log 1234$

$f(x)=\frac{Ax+B}{xC-A}$

$f(f(x))=\frac{A\left(\frac{Ax+B}{xC-A}\right)+B}{C\left(\frac{Ax+B}{CX-A}\right)-A}$

$=\frac{A^2x+AB+xBC-AB}{ACx+BC-ACx+A^2}$

$=\frac{x\left(A^2+BC\right)}{\left(A^2+BC\right)}=x$

$f(f(x))=x$

$f\left(f\left(\frac{4}{x}\right)\right)=\frac{4}{x}$

$f(f(x))+f\left(f\left(\frac{4}{x}\right)\right)$

$AM \geq GM$

$x+\frac{4}{x}\geq 4$

58. The equation $x^2-4x+[x]+3=x[x]$, where $[x]$ denotes the greatest integer function, has :

(A) no solution (B) a unique solution in $(-\infty,\infty)$

(C) a unique solution in $(-\infty, 1)$ (D) exactly two solution in $(-\infty, \infty)$

समीकरण $x^2 - 4x + [x] + 3 = x[x]$, जहाँ $[x]$ महत्तम पूर्णांक फलन है :

(A) का कोई हल नहीं है (B) का $(-\infty, \infty)$ में ठीक एक हल है

(C) का $(-\infty, 1)$ में ठीक एक हल है (D) का $(-\infty, \infty)$ में ठीक दो हल है

Ans. B

Sol. $x^2 - 4x + [x] + 3 = x[x]$

$\Rightarrow x^2 - 4x + 3 = x[x] - [x]$

$\Rightarrow (x-1)(x-3) = [x] . (x-1)$

$\Rightarrow x = 1$ or $x - 3 = [x]$

$\Rightarrow x - [x] = 3$

$\Rightarrow \{x\} = 3$ (Not Possible)

Only one solution $x = 1$ in $(-\infty, \infty)$

59. The number of functions $f : \{1,2,3,4\} \to \{a \in \mathbb{Z} : |a| \le 8\}$ satisfying $f(n) + \frac{1}{n} f(n+1) = 1, \forall n \in \{1,2,3\}$ is

$f(n) + \frac{1}{n} f(n+1) = 1, \forall n \in \{1,2,3\}$ को संतुष्ट करने वाले फलनों $f : \{1,2,3,4\} \to \{a \in \mathbb{Z} : |a| \le 8\}$

की संख्या है

(A) 3 (B) 2 (C) 4 (D) 1

Ans. B

Sol. $nf(n) + f(n+1) = n$

$f(1) + f(2) = 1$

$2f(2) + f(3) = 2$

$3f(3) + f(4) = 3$

$2.(1 - f(1)) + f(3) = 2$

$f(3) = 2f(1)$

$3.2f(1) + f(4) = 3$

$f(4) = 3 - 6\, f(1) \le 8$

$-8 \le f(4) \le 8$

$-8 \le 3 - 6\, f(1) \le 8$

$-11 \le -6f(1) \le 5$

$-\frac{5}{6} \le f(1) \le \frac{11}{6}$

$f(1) = 0, 1$

C–I: $f(1) = 0 , \Rightarrow f(2) = 1$

$f(3) = 0$, $f(4) = 3$

C–II: $f(1) = 1$, $\Rightarrow f(2) = 0$

$\Rightarrow f(3) = 2$, $f(4) = -3$

so '2' such functions.

60. Number of functions $f : \{1, 2, ..., 100\} \to \{0, 1\}$, that assign 1 to exactly one of the positive integers less than or equal to 98 , is equal to _____.

फलनों $f : \{1, 2, ..., 100\} \to \{0, 1\}$, जो 98 से छोटे या उसके बराबर धनात्मक पूर्णांकों में से ठीक एक को 1 पर निरूपित करते हैं, की संख्या है _____.

Ans. 392

Sol. No. of ways to connect one of

$\{1, 2, 3, ...98\}$ to 1 : ${}^{98}C_1 = 98$

No. of ways to connect 99 and 100 = 2×2

Required ans $98 \times 4 = 392$

61. Consider the function $f : R \to R$ defined by $f(x) = \frac{2x}{\sqrt{1+9x^2}}$. If the composition of $f, \underbrace{(fofofo.....f)}_{10 \text{ times}}(x) = \frac{2^{10}x}{\sqrt{1+9\alpha x^2}}$, then the vlaue of $\sqrt{3\alpha+1}$ is equal to _____.

विचार कीजिए फलन $f : R \to R$, $f(x) = \frac{2x}{\sqrt{1+9x^2}}$ द्वारा परिभाषित है। यदि f का संयोजन $f, \underbrace{(fofofo.....f)}_{10 \text{ times}}(x) = \frac{2^{10}x}{\sqrt{1+9\alpha x^2}}$ है, तो $\sqrt{3\alpha+1}$ का मान बराबर है _____

Ans. 1024

Sol. $f(x) = \frac{2x}{\sqrt{1+9x^2}}$

$$(fof)(x) = \frac{2f(x)}{\sqrt{1+9(f(x))^2}} = \frac{\frac{4x}{\sqrt{1+9x^2}}}{\sqrt{1+9\times\frac{4x^2}{1+9x^2}}} = \frac{4x}{\sqrt{1+45x^2}}$$

$$(fofof)(x)=\frac{4\times\frac{2x}{\sqrt{1+9x^2}}}{\sqrt{1+45\times\frac{4x^2}{1+9x^2}}}=\frac{8x}{\sqrt{1+21\times9x^2}}$$

$$(fofofof)(x)=\frac{16x}{\sqrt{1+85\times9x^2}}$$

$\Rightarrow \alpha$ is 10th term of 1, 5, 21, 85, ...

α is 10th term of $\frac{\left(2^1\right)^2-1}{3},\frac{\left(2^2\right)^2-1}{3},\frac{\left(2^3\right)^2-1}{3},\frac{\left(2^4\right)^2-1}{3},\ldots$

$$\Rightarrow \alpha=\frac{\left(2^{10}\right)^2-1}{3}$$

$$\Rightarrow \sqrt{3\alpha+1}=2^{10}=1024$$

62. If $S=\{a\in R:|2a-1|=3[a]+2\{a\}\}$, where [t] deonts the greatest integer less than or equal to t and {t} represents the fractional part t, then $72\sum_{a\in S}$ a is equal to _____.

यदि $S=\{a\in R:|2a-1|=3[a]+2\{a\}\}$ है, जहाँ [t] महत्तम पूर्णांक फलन है तथा {t}, t के भिन्नात्मक भाग को दर्शाता है, तो $72\sum_{a\in S}$ a बराबर है _____

Ans. 18

Sol. $|2a - 1| = 3[a] + 2\{a\}$

Case-I : $a \geq \frac{1}{2}$

$2a - 1 = [a] + 2a$ $\quad (\because a = [a] + \{a\})$

$[a] = -1$

$a \in [-1, 0)$ but $a \geq \frac{1}{2}$, $\Rightarrow$ No value of a

Case-II: $a < \frac{1}{2}$

$1 - 2a = [a] + 2a$

$1 = [a] + 4a$

$1 = 5[a] + 4\{a\}$

$4\{a\} = 1 - 5[a]$

$0 \le 4\{a\} < 4$

$0 \le 1 - 5[a] < 4$

$\frac{1}{3} \ge [a] \ge -\frac{3}{5} \Rightarrow [a] = 0 \Rightarrow \{a\} = \frac{1}{4}$

$\Rightarrow a = \frac{1}{4}$

Now, $72\sum a = 72 \times \frac{1}{4} = 18$

63. If a function f satisfies $f(m+n) = f(m) + f(n)$ for all $m, n \in N$ and $f(1) = 1$, then the largest natural number λ such that $\sum_{k=1}^{2022} f(\lambda + k) \le (2022)^2$ is equal to _____.

यदि सभी $m, n \in N$ के लिए, $f(m + n) = f(m) + f(n)$ है तथा $f(1) = 1$ है, के अधिकतम धन पूर्णांक λ जिसके लिए $\sum_{k=1}^{2022} f(\lambda + k) \le (2022)^2$ है, है _____

Ans. 1010

Sol. $f(m+n) = f(m) + f(n)$

$\Rightarrow f(x) = kx$

$\Rightarrow f(1) = 1$

$\Rightarrow k = 1$

$f(x) = x$

Now

$\sum_{k=1}^{2022} f(\lambda + k) \le (2022)^2$

$\Rightarrow \sum_{k=1}^{2022} (\lambda + k) \le (2022)^2$

$\Rightarrow 2022\lambda + \frac{2022 \times 2023}{2} \le (2022)^2$

$\Rightarrow \lambda \le 2022 - \frac{2023}{2}$

$\Rightarrow \lambda \le 1010.5$

64. Let $A = \{(x, y) : 2x + 3y = 23, x, y \in N\}$ and $B = \{x : (x, y) \in A\}$. Then the number of one-one functions from

A to B is equal to _____.

माना $A=\{(x,y):2x+3y=23,x,y\in N\}$ तथा $B=\{x:(x,y)\in A\}$ हैं। तो A से B के एकैकी फलनों की संख्या है _____.

Ans. 24

Sol. $2x + 3y = 23$

$x = 1 \quad y = 7$

$x = 4 \quad y = 5$

$x = 7 \quad y = 3$

$x = 10 \quad y = 1$

A	B
(1, 7)	1
(4, 5)	4
(7, 3)	7
(10, 1)	10

The number of one-one functions from A to B is equal to 4!

65. Let A = {1, 2, 3, ..., 7} and let P (A) denote the power set of A. If the number of functions $f: A \to P(A)$ such that $a \in f(a)$, $\forall a \in A$, is m^n, m and $n \in N$ and m is least, then m + n is equal to _____.

माना A = {1, 2, 3, ..., 7} है तथा माना समुच्चय A का घात (Power) समुच्चय P(A) है। यदि फलनों $f: A \to P(A)$, जिनके लिए $a \in f(a)$, $\forall a \in A$ है, की संख्या m^n, m, $n \in N$ हैं तथा m न्यूनतम है, तो m + n बराबर है _____।

Ans. 44

Sol. $n(P(A)) = 2^7$

$a \in f(a)$

$\Rightarrow$ f(a) will have (2^6) different subsets having a in them as choice.

$\Rightarrow (2^6)^7 = 2^{42}$ function

$m^n = 2^{42}$

$\Rightarrow m + n = 44$

66. Let R = {a, b, c, d, e} and S = {1, 2, 3, 4}. Total number of onto functions $f: R \to S$ such that $f(a) \neq 1$, is equal to _____.

माना R = {a, b, c, d, e} तथा S = {1, 2, 3, 4} हैं। आच्छादक फलनों $f : R \to S$ जिनके लिए $f(a) \neq 1$ है, की कुल संख्या है :

Ans. 180

Sol. Total onto function

$$\frac{\lfloor 5}{\lfloor 3 \lfloor 2} \times \lfloor 4 = 240$$

Now when f(a) = 1

$$\lfloor 4 + \frac{\lfloor 4}{\lfloor 2 \lfloor 2} \times \lfloor 3 = 24 + 36 = 60$$

so required $f^n = 240 - 60 = 180$

67. Let m and n be the numbers of real roots of the quadratic equations $x^2 - 12x + [x] + 31 = 0$ and $x^2 - 5|x + 2| - 4 = 0$ respectively, where [x] denotes the greatest integer $\leq x$. Then $m^2 + mn + n^2$ is equal to _____.

माना समीकरणों $x^2 - 12x + [x] + 31 = 0$ तथा $x^2 - 5|x + 2| - 4 = 0$ के वास्तविक मूलों की संख्या क्रमशः m तथा n है, जहाँ [x] महत्तम पूर्णांक $\leq x$ है। तो $m^2 + mn + n^2$ बराबर है :

Ans. 9

Sol. $x^2 - 12x + [x] + 31 = 0$

$$\underbrace{x^2 - 12x + 31}_{\geq -5} + [x] = 0$$

It could have its solution in [5, 6] but it does not exist as at x = 5 and 6

LHS = 1

m = 0

$x^2 - 5|x + 2| - 4 = 0$

Case 1

$x \geq -2$

$x^2 - 5(x + 2) - 4 = 0$

$x^2 - 5x - 14 = 0 \Rightarrow x = 7, -2$

Case 2

$x < -2$

$x^2 + 5x + 10 - 4 = 0$

$x = -2, -3$

3 solution i.e., $x = -3, -2, 7$

n = 3

68. Let A = {1, 2, 3, 4, 5} and B = {1, 2, 3, 4, 5, 6}. Then the number of functions $f : A \to B$ satisfying $f(1) + f(2) = f(4) - 1$ is equal to _____.

माना A = {1, 2, 3, 4, 5} तथा B = {1, 2, 3, 4, 5, 6} हैं। तो $f(1) + f(2) = f(4) - 1$ को संतुष्ट करने वाले फलनों $f : A \to B$ की संख्या है _____

Ans. 360

Sol. $f(1)+f(2)+1=f(4)$

$\Rightarrow f(1)+f(2)\le 5$

Let $f(4)=6$, then $f(1)+f(2)=5$

So (f_1, f_2) may be (1, 4), (4, 1), (2, 3), (3, 2)

Total ways = 4 . 1 . 6 . 6 = 144

Let f(4) = 5, f(1) + f(2) = 4

so (f_1, f_2) may be (1, 3), (3, 1), (2, 2)

Total ways = 3 . 1 . 6 . 6 = 108

Let f(4) = 4, f(1) + f(2) = 3

so (f_1, f_2) may be (1, 2), (2, 1)

Total ways = 2 . 1 . 6 . 6 = 72

Let f(4) = 3, f(1) + f(2) = 2

Only way (1, 1)

Total ways = 1 . 1 . 36 = 36

69. Let $[\alpha]$ denote the greatest integer $\le \alpha$. Then $\left[\sqrt{1}\right]+\left[\sqrt{2}\right]+\left[\sqrt{3}\right]+\ldots+\left[\sqrt{120}\right]$ is equal to _____.

माना $[\alpha]$ महत्तम पूर्णांक $\le \alpha$ है। तो $\left[\sqrt{1}\right]+\left[\sqrt{2}\right]+\left[\sqrt{3}\right]+\ldots+\left[\sqrt{120}\right]$ बराबर है _____

Ans. 825

Sol. $S=\left[\sqrt{1}\right]+\left[\sqrt{2}\right]+\left[\sqrt{3}\right]+\ldots.+\left[\sqrt{120}\right]$

$\left[\sqrt{1}\right]\rightarrow\left[\sqrt{3}\right]=1\times 3$

$\left[\sqrt{4}\right]\rightarrow\left[\sqrt{8}\right]=2\times 5$

$\left[\sqrt{9}\right]\rightarrow\left[\sqrt{15}\right]=3\times 7$

$\vdots$

$\left[\sqrt{100}\right]\rightarrow\left[\sqrt{120}\right]=10\times 21$

$S=1\times 3+2\times 5+3\times 7+\ldots.+10\times 21$

$=\sum_{r=1}^{10} r(2r+1)$

$=2\sum_{r=1}^{10} r^2+\sum_{r=1}^{10} r$

$=\frac{2\times 10\times 11\times 21}{6}+\frac{10\times 11}{2}$

= 770 + 55
= 825

70. Let $f(x)=\sum_{k=1}^{10} k\, x^k, x \in \mathbb{R}$. If $2f(2)+f'(2)=119(2)^n+1$ then n is equal to _____.

माना $f(x)=\sum_{k=1}^{10} k\, x^k, x \in \mathbb{R}$ है। यदि $2f(2)+f'(2)=119(2)^n+1$ है, तो n बराबर है _____

Ans. 10

Sol. $f(x)=\sum_{k=1}^{10} kx^k$

$f(x) = x + 2x^2 + ... + 10\,x^{10}$
$f(x) \cdot x = x^2 + 2x^3 + ... + 9x^{10} + 10x^{11}$
$f(x)\,(1-x) = x + x^2 + x^3 + ... + x^{10} - 10\,x^{11}$

$$f(x)=\frac{x\left(1-x^{10}\right)}{(1-x)^2}-\frac{10x^{11}}{(1-x)}$$

$$f(x)=\frac{x-x^{11}-10x^{11}+10x^{12}}{(1-x)^2} \Rightarrow \frac{10x^{12}-11x^{11}+x}{(1-x)^2}$$

Hence $2f(2)+f'(2) = 119.2^{10} + 1$
$\Rightarrow$ So, n = 10